

Advances in Retrieval-Augmented Generation for Scientific and Technical QA: A Review

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ARTICLE INFO

Article History:

Accepted : 01 April 2026

Published: 03 May 2026

Publication Issue

Volume 12, Issue 3

May-June-2026

Page Number

31-42

ABSTRACT

Retrieval-Augmented Generation (RAG) works by using big language models together with external knowledge retrieval systems to increase quality and context of text it generates. Traditional LLMs are limited because the information they provide is drawn from static data, making it hard for them to keep up. RAG deals with this problem by finding appropriate documents or segments from an external knowledge source and sending them to the generative model for processing. This approach to fetching information allows RAG to respond with more context, better facts and greater understanding. Usually, the process includes a retriever that selects the best text groups from the database related to the query, followed by a generator that combines the input and the selected content to produce coherent text. RAG has performed well in processes like open-domain question answering, summarization and analyzing documents in an enterprise environment, mainly in domains such as healthcare, law and finance. Because it does not need retraining to process fresh data, it's very efficient and suitable for real-world situations. Building RAG into NLP is important as it helps make language models more understandable, explanatory and knowledgeable.

Keywords—Retrieval-Augmented Generation, Large Language Models, Natural Language Processing, Information Retrieval, Question Answering, Document Understanding, Contextual Generation

Introduction

Retrieval-Augmented Generation (RAG) is a major approach in NLP that helps unite information retrieval and the process of text production. RAG benefits from combining knowledge-finding retrievers and large language models (LLMs) for the generation of responses, which helps with factual

correctness, and avoids hallucination [1][2] and makes the answers more tailored to the specific user [3]. This kind of architecture works very well in domains where the use of up-to-date, contextual information is important. Evaluating RAG systems ensures that both retrieval methods and generative models are integrated appropriately[4]. The standard evaluation

methods, which use complete system tests to compare results against actual correct results, have significant weaknesses because they cannot evaluate retrieval performance in isolation. RAG operates by retrieving relevant documents from large corpora and injecting this information into the prompt or context for the language model, thereby grounding its responses in factual evidence. The recent study demonstrates that RAG achieves effective performance results throughout multiple natural language processing tasks, which include open-domain question answering and domain-specific applications[5].

The assessment of RAG-based systems for their ability to synthesize complex scientific information became more effective through QA testing. Paper QA demonstrated strong performance surpassing both academic baselines and commercial QA tools y q` [6], and approaching human-level accuracy while remaining cost-efficient. The results demonstrate that RAG provides essential support for applications that require a complete understanding of specialized research materials. QA improved its ability to identify its knowledge limitations, which helps prevent the spread of false information. RAG needs evaluation methods that extend beyond its current end-to-end metrics because this development will enable RAG to gain acceptance in scientific and technical fields [7]. QA is a classic task in NLP that researchers have approached using three methods: rule-based systems, pattern-mining techniques, and deep learning solutions. RAG presents a compelling advancement by leveraging external retrieval to supplement LLMs with relevant, timely knowledge[8]. RAG integration proves especially advantageous for specialized QA tasks, including those in biomedical and scientific fields.

Structure of the Paper

This paper is structured as follows: Section I introduces the motivation for using RAG in scientific and technical question answering (QA). Section II explains the foundational concepts of RAG, focusing on its retriever and generator components. Section III

explores the adaptation of RAG for domain-specific scientific and technical QA. Section IV presents applications and case studies in fields such as biomedicine, law, and engineering. Section V present literature review summary on Retrieval-Augmented Generation for Scientific and Technical QA, and Section VI discusses conclusion and future directions.

Fundamentals Study of Retrieval-Augmented Generation (RAG)

RAG is a brand-new mixed system that claims to be able to fix the issues with pure generative models. There is a system that gets relevant papers or information from an outside source of knowledge. This is then used by a generation module to make text that looks like it was written by a person [9][10]. RAG models use this method to produce natural text and ensure it is backed by current, relevant data, as shown in Figure 1.

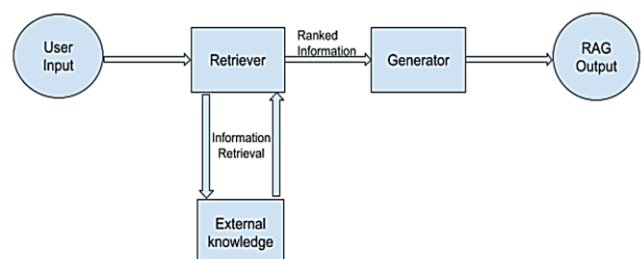


Figure 1. A Basic Flow of the RAG System Along with Its Components

In a component of RAG, the retrieval module uses dense vectors to find and choose documents from Wikipedia and private databases that are related to the input. Once these papers are gathered, they are sent to the generator module, which is usually made with a transformer model, so it can use the information to come up with answers. This method lowers the chance of dreams and makes sure the text fits with the situation. Before they were popular, RAG models were used for open-domain question answering, robots, and making personalized suggestions.

Core components and Architectural of RAG system

RAG works by adding the capabilities of external information search to large language models. The process RAG relies on is for a language model to use stored knowledge, obtain new information and then create a response for the user [11]. The main distinction between RAG and traditional methods in language model adaptation is that RAG adaptations happen as learning continues, rather than being fixed as in fine-tuning. In comparison, RAG enables the model to respond continuously and easily to shifts in its knowledge base, whereas fine-tuning requires retraining each time. RAG enables the model to update its knowledge continuously without retraining, which is vital in areas where data can quickly become obsolete.

Basically, the RAG system uses two components that work side-by-side to deliver accurate, timely and contextual information [12]:

1) Retrieval System (External Knowledge Sources)

In a RAG system, the retriever plays a crucial role in bringing back useful documents from the external corpus. Retrieval works to guarantee the information in the model is accurate [13]. Many approaches for retrieval are available, from the simple BM25 to the complex DPR system.

- The retrieval system is designed to retrieve necessary facts from resources found outside the system.
- It gathers and presents ones that go along with the user's query, helping the AI model obtain the most relevant and recent data.
- As one of the oldest information retrieval algorithms, BM25 determines a document's relevance by ranking it using TF-IDF. Although existing for many years, BM25 is still considered a benchmark for many modern retrieval systems used in RAG models.

2) Generative Model (Text Synthesis)

In RAG systems, the generator uses the input and information it retrieves to produce the final output. The generator uses what is found by the retrieval

component to form responses that are logical and fit the situation.

- This approach creates natural-sounding text using what the user asks and what is found relevant.
- As a result, its responses are clear, natural and in the right context, also relying on the mentioned information to ensure accuracy.
- Text-to-Text Transfer Transformer (T5) is used by many RAG systems to make text. The way T5 works is flexible, it sees every NLP job as a text-to-text task. This consistent framework lets T5 be fine-tuned for many jobs, such as answering questions, summarizing, and coming up with dialogues.
- A well-known generation model used in RAG systems is BART, which stands for "Bidirectional and Auto-Regressive Transformer." BART works especially well for jobs like summarization and open-domain question answering that require creating text from noisy inputs.

Challenges and Limitations of Retrieval-Augmented Generation (RAG)

The RAG systems are generally Intrigate. This particular setup, on the other hand, with randomly splitting text into chunks and basic top-K retrieval based on a similarity measure, could be called "naive RAG." Many problems exist with this method, including misaligned text chunks and difficulty remembering important sections. Additionally, issues may arise during the generation phase, such as answers that are repetitive or don't make sense, answers that aren't based on the given context, and issues with balancing the value of different retrieved passages. stated that some issues with RAG systems happen because the retrieval and generation steps are handled separately. This is expanded upon by noting that humans and LLMs have different preferences regarding the style of retrieved texts and how they are presented to the LLM. They also say that retrieved contexts are usually changed to suit the tastes of humans rather than LLMs [14]. As a result, they say that a solution to this problem must be found.

The purpose of this section is to talk about the difficulties and problems that RAG faces today:

- Scaling RAG models is a major issue that needs to be addressed. Handling large and changing datasets is possible with efficient retrieval tools that access external databases.
- Ensuring that the information found in documents is recent and of good quality is still a big concern. Sometimes, retrieval models give information that is unrelated to what the user wants and can leave the output less accurate.
- RAG systems may end up with bias as a result of the biases found in the data sets used. Models that rely on retrieval could transfer biases present in learned knowledge, resulting in biased generated material.
- It can be tough for RAG models to unify the collected details into meaningful and logical sentences for a given context. Passages from the GPT model are not always consistent with what is recorded in the reference material, leading to some incorrect or made-up information in the answer.

RAG models are often thought of as "black boxes" because it's not always clear how retrieval affects production. Improving how easy it is to understand these models is important for building trust, especially in serious situations

Generative Models of Retrieval-Augmented Generation (RAG)

The RAG model, which was created by, combines retrieval processes with generative models to make responses that are more relevant to the situation and make more sense[15]. The RAG model improves on many of the problems with standard search engines by finding relevant documents and using that information to generate answers [16]. These results

show how well RAG works in different NLP tasks, such as answering questions and summarizing in key points[17].

- Information retrieval (IR) systems focus on retrieving the most relevant documents but do not generate responses.
- Although often employed, models like BM25 and TF-IDF have trouble recognizing natural language.
- Integrates the flexibility of generative models with retrieval-based accuracy.
- Research indicates that RAG improves contextual relevance and information retrieval accuracy.
- Research has shown that RAG outperforms standard search engines in terms of user satisfaction and query resolution efficiency.

Transformers on Generators in RAG

The system requires the generator to create final responses by combining information obtained through retrieval with the current query data. The generator creates authentic and suitable answers by using all available knowledge after the retrieval process reaches completion. The Large Language Model (LLM) powers the generator which creates text that meets three standards of clarity, accuracy and relevance to the query [18][19].BERT, GPT and T5 models require study because they enable translation technology development according to [20] since they serve as fundamental components of natural language processing and support multiple translation use cases. BERT dominates contextual embedding, GPT excels at producing coherent written content and T5 provides a unified framework for handling translation assignments through its text-to-text conversion system. Table I provides an overview of these models, listing their main points and the benefits they bring to translation services and tasks[21]:

TABLE I. COMPARATIVE ANALYSIS OF BERT, GPT, AND T5 FOR MACHINE TRANSLATION

Model	Key Features	Translation Benefits
BERT	• Bidirectional encoder that understands the context of words in both directions. • Captures relationships between words in a sentence.	• Enables handling of complex sentence structures and ambiguous phrases. • Improves semantic understanding for accurate translations. • Useful for fine-tuning translation models with contextual embeddings.
GPT	• Generative model that produces coherent, context-aware text. • Uses autoregressive decoding (predicts next word from previous context). • Strong inventive generation capabilities.	• Produces fluent, natural translations. • Maintains coherence and contextual relevance. • Enables expressive, culturally appropriate translations.
T5	• Treats all NLP tasks as text-to-text problems. • Learns task-agnostic text representations. • Highly adaptable via fine-tuning.	• Simplifies translation pipeline across language pairs. • Enables efficient transfer learning for specific tasks. • Supports domain-specific or stylistic translations (e.g., technical, literary).

The BERT, GPT, and T5 systems use different methods to perform machine translation. BERT achieves its strong ability to understand complex sentences through its bidirectional encoding system. The autoregressive generation of GPT enables it to create translations that are both fluent and coherent. T5 unifies NLP tasks under a text-to-text framework, offering flexibility and strong transfer learning across language pairs [22]. Each model has unique strengths suited to specific translation needs semantic accuracy (BERT), fluency (GPT), and adaptability (T5).

Target Data Fine-tuning and Domain Drift Boosting

Squad, a high-quality and rich dataset, is used to match the model to the open domain development environment. When the text encoder has been fine-tuned, target data fine-tuning generally follows. As a result, this approach might not work well with a dataset that is very different from Squad in terms of both grammar and language. They are doing this because the information moves around a lot. Furthermore, LMs are known to be inaccurate and

have a tendency to fit the data too well [23]. Avoid this effect because it works better in low-budget settings. It is considered merging the Squad and Target QA datasets as a way to fix this problem. This would improve the stability of the fine-tuning process and stop the bad forgetting that happens a lot during QA fine-tuning. When the merged fine-tuning method uses the original hyperparameters used in Squad fine-tuning, it can get the most out of them and avoid making mistakes that can happen when looking through a lot of them [24]. A lot of people use samples as general samples, but it's expensive and unlikely that you'll be able to get high-quality training datasets the size of Squad for every subject. Instead, they choose to under- or over-sample the datasets to focus on the domain-specific data. This came from ideas in classification with unbalanced classes.

Applications Of Scientific And Technical Question Answer (QA)

RAG has emerged as a powerful paradigm in tackling complex question answering (QA) tasks across scientific and technical domains [25]. Scientific and technical question answering needs exact answers from specialized vocabulary knowledge and complete documentation resources which differ from standard question answering. RAG models address these challenges by retrieving relevant supporting documents and generating context-aware responses that improve answer accuracy and trustworthiness. RAG systems have been used in biomedical research to conduct literature-based discovery and clinical decision support through the analysis of Bio ASQ and PubMed QA datasets[26][27][28]. RAG models enable patent-related question answering and engineering tasks through their ability to analyze technical documents and perform prior art research while retrieving essential patent information and design materials. RAG applications show researchers and professionals how to use information better while scientific workflows with their potential to change research practices. The field still needs to handle three main research problems which include domain adaptation issues, hallucination problems and retrieval quality challenges which establish future research directions as shown in (Figure 2).

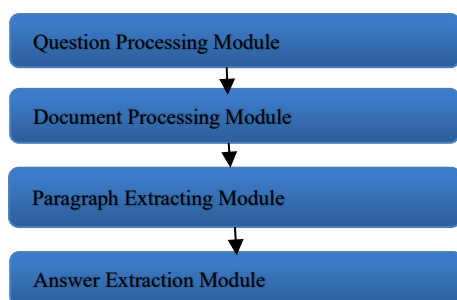


Figure 2. QA System

Figure 2 shows how a standard question-answering (QA) system is put together. The system consists of four components which establish a top-down pipeline

system. The Question Processing Module begins the process by analyzing the input question to organize it into different categories. After this comes the Document Processing Module, whose job it is to get important documents from the corpus. The Paragraph Extracting Module then chooses the parts that are most likely to have the answer, which narrows the search. The Answer Extraction Module takes the selected text to extract the specific answer from it. The modular architecture serves as the standard design pattern for information retrieval-based QA systems because it enables systems to scale their capacity while maintaining their ability to process natural language questions.

Biomedical Questions Answers (QA)

Multiple biomedical knowledge bases have been developed to assist biomedical question answering, enabling users to retrieve answers via natural language queries converted into RDF queries for knowledge base searches. This was called the Explicit Knowledge Querying approach for KB BQA [29]. More recent studies seek to integrate biomedical knowledge into neural models through pre-training or reasoning modules. They see these as the Implicit Knowledge Integration method for KB BQA [30].

KBs: The Biomedical KBs are like databases that list biomedical things and how they relate to each other. These can usually be kept in RDF subject-predicate-object triples. Building biomedical knowledge bases has taken a lot of work. These databases include ontologies such as Medical Subject Headings (MeSH) for biomedical topics, Gene Ontology for genetic terms, International Classification of Diseases (ICD) for diseases, and Systematized Nomenclature of Medicine Clinical Terms for medical terms. A meta-thesaurus called the Unified Medical Language System (UMLS)⁸ is made up of almost 200 different biological keywords, such as Mesh and ICD.

Issues of QAS Framework Based on RAG

There are a number of problems with the way a question-answering (QA) method is set up in RAG [31]:

- **Ambiguity:** Natural language questions aren't always clear and can be interpreted in different ways. This makes it hard for the QA system to figure out what the question is really asking and get the right answer.
- **Vocabulary:** The QA system is likely to face difficulties making sense of unusual terminology or informal expressions in questions.
- **Lack of context:** Sometimes, the QA system does not have enough information to correctly answer the question.
- **Complex or open-ended questions:** Some questions that are hard to classify may be outside the ability of the QA system to address.
- **Limited dataset:** The QA system might not give an answer when the necessary information isn't in its training data.
- **Biases:** An unfair QA system may exist if the dataset it was trained on contains biased information.
- **Lack of evaluation:** Implementing the QA system before it is properly evaluated might result in it not operating optimally in real use.
- **Lack of scalability:** The QA system could struggle with lots of questions and information.

One way to address these issues is to make sure QA systems use powerful NLP techniques and are trained using vast and varied data that is not biased. The QA process should also be checked and changed from time to time to raise its effectiveness and fix any problems.

Question Answering Technology

Natural language questions can be answered exactly with this technology. Question answering systems are tools that use this technology. Customers can get clear answers to their questions by just asking the system; the answers are shown right away, and customers don't have to read any papers. Different kinds of questions are answered by question answering systems [32]. Systems are split into two groups based on the kinds of questions they can answer. One type is a factoid question answering system, which uses

words or short sentences to answer factual questions. The other is a method for answering questions that doesn't use factoids. It gives answers in the form of sentences or paragraphs. For instance, the old method gives the answer "Everest" to the question "What is the tallest mountain in the world?" and "George Washington" to the question "Who was the first president of the United States?" For instance, the second one gives complete answers to meaning questions, "why" questions, and "how" questions. Users could ask any question, so if knew all of them ahead of time, could prepare the right answers and give them when they are needed. In practice, though, user questions are too different to be able to predict. So, question answering systems work the same way people do to find answers: they figure out what the question means, look for related documents, and find answers.

Literature Review

This section presents a literature review on Advances in Retrieval-Augmented Generation for Scientific and Technical QA: A Review. An overview of emerging threats and best practices. Table II offers a succinct overview of the examined research.

Jain, D.M. (2026) incorporating knowledge-aware generating frameworks, retrieval tactics, embedding methods, reranking strategies, and system architecture. RAG has been shown to provide interpretability, multi-hop query answering, and evidence-based reasoning in open-domain question answering applications. The applicability of RAG to real-world scenarios has been demonstrated by its application in conversation systems, domain-specific assistants, scientific summarization, enterprise knowledge systems, medical reasoning systems, and code generators outside of QA. Recent advancements that enhance RAG's capabilities include hybrid retrieval techniques, graph-based augmentation, multimodal integration, and agent-like reasoning[33]. Hindi et al. (2025) a way that RAG is used in the law. It was first explained in brief what the main RAG

steps, methods, techniques, and legal uses are. There was then a quick review of the various information retrieval models, processes, and methods utilized in present law RAG solutions. Second, they talked about the various numeric and qualitative assessment tools, as well as some fresh data sets and standards. Third, they talked about and rated the privacy and ethical problems that come up with legal RAG. They also summed up the different problems and offered a challenge scale based on these problems and how much control the outside world has over RAG. Finally, it talked about some cool things that could be researched in the future to make RAG more useful in the legal field [34].

Tural et al. (2024) emphasize that the integration of RAG architecture with information retrieval systems and LLMs provides more sensitive and accurate solutions in information-intensive tasks. This work highlights how applications in the field of NLP are strengthened by the RAG architecture's capacity to extract information by dynamically utilizing the learnings acquired from vast datasets of LLMs. Advances in NLP have led to the emergence of complex structures such as Large Language Models (LLMs). LLMs are highly successful in understanding the subtleties of language and processing context by being trained on large datasets. However, the difficulties encountered in Information Retrieval (IR) processes have created an awareness that these models are not sufficient on their own [35].

Saha et al. (2024) a new way to build RAG systems that can make QA jobs better from a target corpus. LLMs have changed how writing is analyzed and created, making it sound as if it were written by a person. These models use data that has already been trained and don't get real-time changes unless they are connected to live data tools. RAG improves LLMs by using online resources and databases to generate answers appropriate to the situation. But traditional RAG still has problems with information loss and hallucinations when it deals with huge amounts of data. This method addresses these problems by

transforming corpora into domain-specific datasets. The RAG architecture is built to get answers from the target document, and Question-to-question Inverted Index Matching, or Qumix-RAG, is a new way to improve the system's retrieval process [36].

Omrani et al. (2024) A new mixed RAG framework that combines the Sentence-Window and Parent-Child methods with a creative re-ranking system is designed to make LLMs better at answering queries. This method gets better results from LLMs by using more information, which makes them more accurate and useful while still staying true to the information being handled. Results show that their method can produce answers that make sense in the context and are true to the original text, which makes it the best way to improve LLM responses. The study finds that mixed RAG models can improve the way LLMs and outside data work together. This makes room for more NLP research[37].

Gao *et al.* (2023) It has been shown that, especially for knowledge-intensive tasks, RAG considerably improves answer accuracy and decreases model hallucination. Users can boost confidence in model outputs and confirm the correctness of replies by acknowledging sources. Additionally, it makes domain-specific knowledge introduction and knowledge upgrades easier. One of the most crucial techniques for constructing large language models is RAG, which successfully blends the parameterized information of LLMs with non-parameterized external knowledge bases. In this study, the three RAG development paradigms—Naive RAG, Advanced RAG, and Modular RAG—are summarized in the context of LLMs. The three primary RAG components—retriever, generator, and augmentation methods—as well as the fundamental technologies used in each are then summarized and arranged[38].

Table II provides an overview of the literature on Retrieval-Augmented Generation for Scientific and Technical QA, listing papers, their methods, key findings, challenges, and future directions.

TABLE II. LITERATURE OF REVIEW ON ADVANCES IN RETRIEVAL-AUGMENTED GENERATION FOR SCIENTIFIC AND TECHNICAL QA A REVIEW

Author	Study On	Method	Key Findings	Challenges	Future Directions
Jain, D.M. (2026)	RAG in scientific and technical QA	Hybrid retrieval, embeddings, reranking, knowledge-aware generation	Enhances multi-hop QA, reasoning, and interpretability; applicable across domains	High cost, retrieval inefficiency, domain limitations	Need for scalable, domain-adaptive models and standardized evaluation
Hindi et al. (2025)	RAG in the legal domain	Overview of RAG stages, techniques, legal IR methods, evaluation metrics	Highlights ethical/privacy considerations, and introduces a challenge scale based on failure points and control over knowledge	Ethical constraints, domain-specific benchmarks, accuracy	Develop domain-specific RAG solutions and ethical frameworks
Tural et al. (2024)	Integration of RAG with IR systems and LLMs in NLP	Analysis of LLM and IR synergy	Demonstrates RAG's improved sensitivity and accuracy in information-heavy tasks	Limitations of standalone LLMs, IR inefficiencies	Combine LLMs with adaptive IR systems for dynamic querying
Saha et al. (2024)	Novel RAG architecture for QA	QuIM-RAG (Inverted Index Matching) with domain-specific corpora	Enhances contextually relevant answers, reduces hallucination	Information dilution, hallucinations in large datasets	Expand domain-specific training, real-time RAG updating
Omrani et al. (2024)	Hybrid RAG using Sentence-Window and Parent-Child models	Innovative re-ranking with external knowledge sources	Outperforms existing RAG methods; improves fidelity and relevance	Source material inconsistency, response ranking	Further optimize hybrid RAG with adaptive ranking algorithms
Gao et al. (2023)	RAG paradigms in LLMs	Naive, Advanced, and Modular RAG with retriever-generator framework	Improves accuracy, reduces hallucination, enables knowledge updates and trust via source citation	Complexity in integration, retrieval quality dependency, scalability issues	Need for unified frameworks, efficient retrieval optimization, and better evaluation of modular RAG systems

Conclusion And Future Work

RAG has shown great potential for advancing natural language processing by joining external information searching with powerful language models to generate accurate and thoughtful answers. Because its architecture is modular, it can easily expand and respond to changes in fields where knowledge updates are needed often, for example, medicine, law and science. The use of Paper QA has enabled RAG to lead in question-answering within specific fields by organizing and matching the information obtained. Yet, while RAG systems are helpful, they still struggle to remember relevant documents and often lack a consistent structure due to conflicts between how documents are retrieved and how summaries are generated. They suggest that students should use both retrieval and generation methods and consider adopting more effective chunking strategies. Researchers should focus on strengthening the link between the retrieval and generation components of RAG systems to reduce errors and improve the relevance of the information provided. RAG systems achieve better performance when adaptive chunking, retrieval-aware generation, and user-feedback integration are implemented together. The development of common standards to evaluate answer retrieval will enable a detailed assessment of each system's limitations. RAG deployment in multilingual environments that require immediate processing will enhance its ability to function effectively across various technical and fast-paced environments.

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